

MAE SOT, Thailand

WMO: 483750

Lat: 16.7025N

Lon: 98.5417E

Elev: 648

StdP: 14.36

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.5	59.1	49.1	53.0	72.6	52.1	59.1	73.0	13.6	82.5	12.0	83.2	0.3	10	0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.5	99.0	75.5	97.2	74.9	95.3	74.6	80.0	89.8	79.2	88.8	78.7	88.0	7.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	146.2	83.9	76.9	143.6	83.0	76.4	140.9	82.3	44.2	90.3	43.4	89.2	42.7	88.3	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	11.8	10.0	DB	51.4	101.9	3.6	2.0	48.8	103.4	46.7	104.5	44.7	105.7	42.1	107.2	(4)
			WB	50.0	83.1	3.5	3.1	47.5	85.4	45.5	87.2	43.5	88.9	41.0	91.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.2	74.9	78.2	82.3	86.0	84.3	81.4	79.7	79.7	80.9	81.1	78.5	74.9
	DBStd	4.29	3.51	3.28	2.86	2.56	3.63	2.43	2.16	2.04	2.01	1.86	3.14	4.04
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD50	11011	772	790	1001	1081	1063	942	921	919	928	965	856	773
	CDD65	5538	308	370	536	631	598	492	456	454	478	500	406	309
	CDH74	53457	2804	3982	6320	8016	6583	4371	3412	3321	3984	4455	3576	2633
	CDH80	21887	1122	2003	3466	4422	2942	1266	761	721	1163	1624	1426	971
Wind	WSAvg	2.4	1.8	2.0	2.9	3.5	3.3	2.6	2.4	2.3	2.0	2.0	2.3	2.2
Precipitation	PrecAvg	62.4	0.3	0.2	0.2	1.0	7.3	10.0	13.7	17.0	7.7	4.2	0.6	0.2
	PrecMax	96.5	1.6	0.9	1.1	3.3	15.0	17.7	36.0	32.4	15.4	10.2	2.4	0.6
	PrecMin	44.6	0.0	0.0	0.0	0.0	0.0	4.0	4.8	8.1	4.1	0.9	0.0	0.0
	PrecStd	13.6	0.5	0.4	0.3	1.3	3.5	4.1	7.9	7.3	3.5	2.6	0.7	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	96.3	99.1	101.6	101.9	93.3	90.3	89.9	91.8	92.6	93.1	91.5
		MCWB	71.9	71.2	73.1	75.9	77.1	78.8	78.4	78.3	78.6	77.5	76.3	73.6
	2%	DB	90.1	94.3	97.3	99.7	98.2	90.7	88.2	88.0	89.8	90.9	91.2	89.6
		MCWB	71.2	71.5	72.8	75.9	77.3	78.3	77.8	77.8	78.3	77.4	75.3	73.1
	5%	DB	88.5	92.7	95.9	98.1	95.0	88.5	86.2	86.1	88.1	89.6	89.6	87.8
		MCWB	70.7	71.1	72.6	75.9	77.5	77.8	77.2	77.5	77.9	77.2	74.8	72.4
	10%	DB	86.6	90.8	94.2	96.3	91.8	86.4	84.2	84.1	86.3	88.0	87.8	85.8
		MCWB	70.2	70.7	72.5	75.9	77.5	77.4	76.7	76.8	77.5	76.9	74.1	71.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.7	77.0	78.0	80.7	81.5	80.5	79.7	79.6	80.1	80.2	78.7	77.1
		MCDB	86.7	90.0	92.5	94.8	93.0	89.2	87.3	86.7	88.3	88.8	89.0	87.1
	2%	WB	74.5	75.0	76.5	79.0	79.8	79.2	78.3	78.5	79.1	79.1	77.5	75.5
		MCDB	85.2	89.1	91.3	93.5	91.1	87.4	85.4	85.5	87.2	87.9	87.2	85.5
	5%	WB	73.1	73.7	75.5	78.1	78.9	78.4	77.7	77.9	78.4	78.3	76.6	74.2
		MCDB	83.9	87.0	89.8	92.4	89.7	86.0	84.2	84.2	85.9	86.7	85.7	84.0
	10%	WB	71.8	72.3	74.6	77.3	78.1	77.8	77.1	77.2	77.8	77.6	75.6	72.9
		MCDB	82.5	85.7	88.4	91.3	88.0	84.5	82.8	82.8	84.6	85.3	84.1	82.3
Mean Daily Temperature Range	5% DB	MDBR	25.7	27.9	25.6	21.5	15.0	11.1	9.7	9.8	11.9	15.3	19.4	22.9
		MCDBR	27.2	29.0	27.5	23.9	19.8	13.9	12.6	12.5	14.4	17.3	21.1	23.8
		MCWBR	10.4	9.6	8.5	7.0	5.7	4.9	4.7	4.6	5.1	6.2	7.4	9.0
	5% WB	MCDBR	23.4	25.1	23.5	21.2	16.1	12.4	11.1	11.2	13.3	15.5	18.7	21.5
	MCWBR	10.2	9.4	7.9	6.8	5.6	5.0	4.8	4.8	5.2	6.0	7.0	8.9	
Clear-Sky Solar Irradiance	taub	0.448	0.531	0.651	0.603	0.494	0.444	0.435	0.438	0.456	0.521	0.455	0.445	
	taud	2.177	1.969	1.695	1.868	2.215	2.383	2.410	2.390	2.324	2.087	2.243	2.213	
	Ebn at Noon	265	248	223	235	258	269	272	273	269	248	261	262	
	Edh at Noon	45	58	78	66	46	39	38	39	41	51	42	43	
All-Sky Solar Radiation	RadAvg	1503	1683	1767	1869	1588	1297	1180	1184	1375	1481	1529	1455	
	RadStd	62	66	108	115	156	142	141	105	82	81	83	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.18	N/A	N/A	N/A	N/A

Nomenclature: See separate page